

Coherence-Rupture-Regeneration

A Temporal Grammar, and Two Points of Contact with Differential Flatness

Alexander Sabine

Active Inference Institute

In dialogue with Mounier, Parr & Friston (2026), Entropy 28(1), 87 · offered for discussion

Abstract

Every finite system that persists must accumulate evidence, reach the limit of its current regime, and reconstruct from its own history. Coherence-Rupture-Regeneration (CRR) proposes that these three operations form a candidate temporal grammar for adaptive systems, governed by a single geometric parameter Ω fixed by the topology of the system's statistical manifold. Two one-dimensional symmetry classes — an interval and a circle — yield two thresholds, π and 2π , with a ratio of exactly two and no free parameters. After setting out the framework and its five commitments, this paper puts CRR into dialogue with the differential-flatness account of active inference given by Mounier, Parr and Friston. Both frameworks represent a trajectory as a functional of its history; they differ in that flatness uses a finite collection of temporal derivatives while CRR uses a global integral. Two consequences are examined and computed directly. First, a delay lies outside the span of every finite jet but inside the closure of the memory-kernel family, locating precisely where a history integral becomes necessary rather than optional. Second, on the Bernoulli manifold the coordinate that linearises the natural-gradient flow coincides with the arc-length coordinate in which CRR's clock runs uniformly — suggesting the two frameworks may, in this setting, share a coordinate rather than merely resemble one another. Throughout, the framework's limits are stated as plainly as its aspirations.

Keywords: *active inference · temporal grammar · information geometry · differential flatness · generalised coordinates · memory · Fisher-Rao*

A note on aim and status

This paper has two parts. The first sets out the CRR framework and the five commitments on which it rests; some of this is on firm ground — the information-geometric definition of coherence, the Čencov-fixed parameter, and the parameter-free ratio between two symmetry classes — and some remains a proposal under active development. The second part puts CRR into dialogue with the differential-flatness account of active inference of Mounier, Parr and Friston (hereafter MPF), and here the aim is deliberately modest: to identify genuine points of contact, to distinguish structural analogy from anything stronger, and to state the mathematical questions

that would tell them apart. Every quantitative claim below was computed directly, and each is presented with the limits of what has actually been shown.

Part I. The CRR Framework

CRR models a bounded system through three coupled temporal operations, resting on five commitments that any persistent, finite, adaptive system may be argued to satisfy. The framework's parsimony — a single geometric parameter governing temporal coordination — is the feature that makes it falsifiable, and the commitments are stated first because they determine everything that follows.

The five commitments

1 • Accumulation	A persisting system gathers what reduces its surprise. Coherence is the Fisher-Rao arc length of its trajectory: $C(x,t) = \int \ \dot{x}\ _g d\tau$. Non-Markovian by construction.
2 • Rupture	A finite manifold has finite geodesic extent. When accumulated arc length spans it, the regime is exhausted and reorganises: $\delta(\text{now})$ when $C \cdot \Omega = 1$.
3 • Regeneration	Reconstruction draws on the whole past, weighted by attained coherence: $R = \int \varphi \cdot \exp(C/\Omega) \cdot \Theta(t-\tau) d\tau$. Significance, not recency.
4 • One parameter	$\Omega = 1/\ell$, fixed by Čencov's theorem, never fitted. It sets rupture rate, memory depth, and boundary rigidity at once. Observable: $CV = \Omega/2$.
5 • The crossing	Whatever the substrate, the rupture event is Z_2 : one binary commitment. Determinate within, a fair choice at the cut, open possibility beyond. This pins the thresholds.

1. Coherence: accumulation as arc length

A system that persists must accumulate the patterns that reduce its surprise. CRR formalises this as coherence — the Fisher-Rao arc length the system travels along its information-geometric trajectory:

$$C(x, t) = \int_0^t \|\dot{x}\|_g d\tau, \quad \|\dot{x}\|_g = \sqrt{(g_{ij} \dot{x}^i \dot{x}^j)},$$

where g is the Fisher information metric. The little g is the ruler: it converts a raw change of parameters into a distance measured in *distinguishable states*, weighting each direction by how much statistical difference a step in it makes. The square root sits inside the integral deliberately — it is what makes C the arc length, the one functional invariant under reparameterisation of time, rather than the energy functional $\frac{1}{2} \int g \dot{x}^2 d\tau$, which is not. Because arc length on a statistical manifold is dimensionless, C is dimensionless, as the rupture condition below requires. The integral runs over the whole history of the current regime, which makes CRR non-Markovian at the process level even within a Markovian generative model.

Read through the Cramér-Rao bound, C has a concrete meaning: for an efficient estimator one unit of Fisher-Rao arc length corresponds to one standard error, so C is a running tally of *just-noticeable differences* — the count of resolvable states the

system has passed through. Under arc-length (unit-speed) parameterisation the speed is identically one and $C(\tau) = \tau$: in its own natural clock, a system's accumulated distinguishability is its elapsed time. This is the sense in which CRR supplies a clock that the Free Energy Principle presupposes but does not itself formalise.

2. Rupture: exhaustion of geodesic extent

No finite system accumulates without bound. A statistical manifold has a finite geodesic extent ℓ , and a system moving along it must eventually span it. That moment is rupture:

$$\delta(now) \text{ when } C \cdot \Omega = 1 ,$$

represented by a Dirac delta of zero temporal width carrying unit mass on the boundary between the accumulated past and the reconstructed future. The condition marks the point at which accumulated arc length equals the manifold's full extent: every distinguishable state available to the current regime has been traversed. It is best described as **exhaustion of geodesic extent** rather than as saturation of an estimator, since in discrete systems rupture is a threshold-crossing — a heartbeat, a breath turning, a neuron firing — and the accumulated coherence overshoots the bound slightly rather than reaching it exactly.

3. Regeneration: significance, not recency

After rupture the system rebuilds from the only resource it has, its own history, weighted by the coherence each past moment attained:

$$R(x, t) = \int \varphi(x, \tau) \cdot \exp(C(x, \tau)/\Omega) \cdot \Theta(t - \tau) d\tau ,$$

where φ is the field of past states available for reconstitution and the Heaviside step enforces causal order. The kernel is Boltzmann in form — C plays the role of energy and Ω the role of temperature — which is the maximum-entropy weighting under a constraint on the expected value of C . History therefore enters by significance rather than recency: a high-coherence moment from long ago outweighs a recent low-coherence one. The dynamic range of this kernel over one cycle is $\exp(\ell^2)$, which is modest for the interval and very large for the circle; in practice regeneration is best computed in log space.

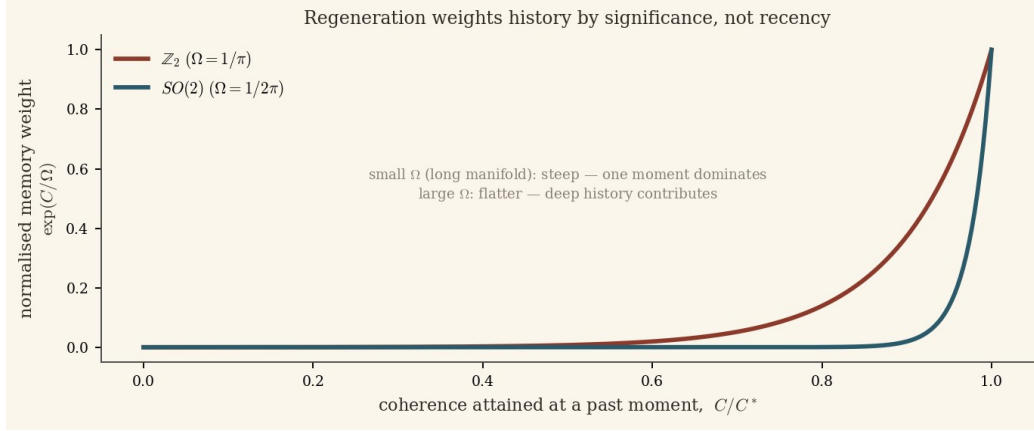


Figure 1. The regeneration kernel $\exp(C/\Omega)$, normalised, for the two symmetry classes. Both weight past moments by attained coherence rather than by recency; the smaller Ω (the longer manifold) gives the steeper kernel, so that a system with small Ω reconstitutes a few dominant moments (rigidity) while one with large Ω draws on a flatter, deeper history (flexibility).

4. One parameter, fixed by geometry

The five commitments leave a single degree of freedom, the value of Ω , and its whole force is that it is not fitted. For any system whose internal states parametrise probability distributions — which under the FEP includes any system with a Markov blanket — the arena of accumulation is a statistical manifold with a definite geometry. Čencov's uniqueness theorem constrains the metric to be the Fisher metric, and the geodesic structure of that metric fixes the maximal arc length. The capacity becomes a geometric invariant,

$$\Omega = 1 / \ell ,$$

the manifold's fractional resolution: the reciprocal of the number of just-noticeable differences that span it. Small Ω means a long manifold — many resolvable states, a sharply peaked memory kernel, rare and deep ruptures, a rigid boundary. Large Ω means the reverse. It is worth being explicit that Ω is not a variance and not an inverse precision; in the CRR-FEP correspondence precision maps to the product $(1/\Omega) \cdot \exp(C/\Omega)$, a topological factor fixed by geometry and a dynamical factor growing with evidence, so $1/\Omega$ is a geometry-fixed baseline that contributes to precision rather than equalling it. Treating Ω as a fitted variance would forfeit exactly the property that gives the framework predictive content.

5. The rupture event, and the two classes

Whatever manifold coherence accumulates on, the rupture *event* is Z_2 : a single unit-mass commitment, this regime or the next. This is because a rupture is a boundary-crossing, and a crossing has exactly two sides regardless of the shape being traversed. It gives the temporal sibling of the Markov blanket — where the spatial blanket separates internal from external states, the Dirac delta separates the accumulated past C (inside) from the reconstructed future R (outside). It also resolves an apparent tension: from within, the state at rupture is maximally determinate (its

entropy vanishes and its Fisher information diverges); the crossing itself is a fair binary, one bit; and what it releases into the future is maximum possibility. The manifold ends because determinacy is complete, and the exhaustion of description is the release of possibility.

Two one-dimensional symmetry classes fix ℓ . The Bernoulli manifold, parametrised by p , is an interval whose Fisher-Rao diameter is π ; a bistable system exhausts its configuration by crossing from one basin to the other. The circle S^1 has circumference 2π ; a rotational system exhausts its configuration by completing one cycle. The ratio between the classes is exactly two — a topological invariant, independent of any physical parameter — and the observable that follows, the one on which the framework is most falsifiable, is a coefficient of variation of inter-rupture intervals equal to $\Omega/2$.

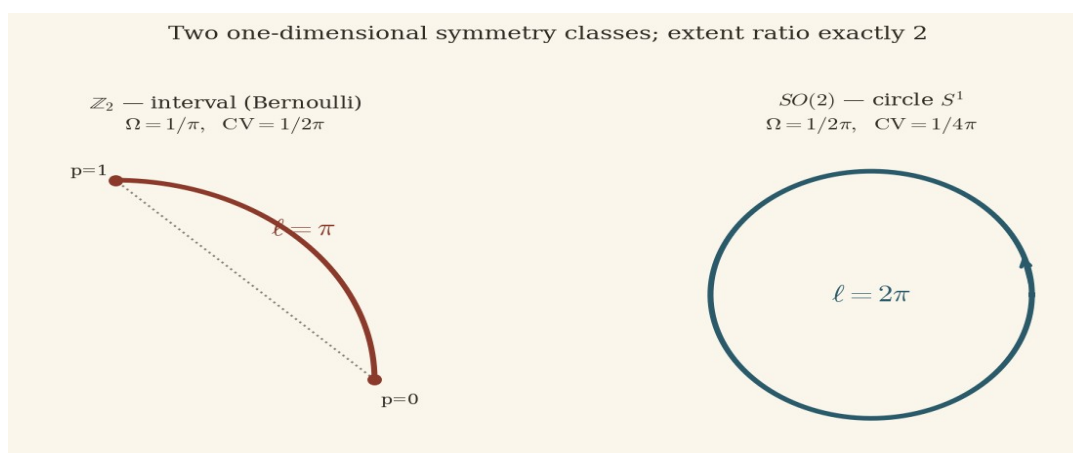


Figure 2. The two symmetry classes as one-dimensional manifolds. Left: the Bernoulli interval, Fisher-Rao extent π (drawn as the quarter-arc of a radius-2 circle, whose length is π). Right: the circle, circumference 2π . $\mathbb{Z}_2$ gives $\Omega = 1/\pi$ and $CV = 1/2\pi \approx 0.159$; $SO(2)$ gives $\Omega = 1/2\pi$ and $CV = 1/4\pi \approx 0.080$; the extent ratio is exactly 2.

A word on the domain of the CV prediction, since it bears on Part II. $CV = \Omega/2$ is a claim about *clocklike* (sub-Poisson) processes, those that pass a burstiness gate; respiration and healthy cardiac rhythms qualify, and it is here that the parameter-free prediction is tested. Noise-dominated processes — whose interval distributions are ex-Gaussian, with a heavy exponential tail — fall into a separate class for which CRR predicts that no clean CV can be recovered. The framework thus carries its own eligibility criterion, and applying the CV prediction outside it would be a misuse rather than a test.

Part II. Two Points of Contact with Differential Flatness

MPF describe active inference and differential flatness as, at first sight, *quite distant frameworks* — the one driving a trajectory-tracking error to zero, the other minimising variational free energy — and then show that the tracking error is itself a form of surprise. Their emphasis falls on the pathwise properties of flat systems: a differentially flat system admits flat outputs such that the state and input can be written as functions of those outputs and finitely many of their time derivatives. This is what ties flatness to the generalised coordinates of motion used in continuous-time active inference — both represent a trajectory by a finite collection of temporal derivatives at the present instant. A flat representation is, in that precise sense, local in time.

CRR's regeneration is the mirror image: a global history functional, an integral over the entire past with a non-trivial memory kernel. So the two frameworks meet on a shared question — how a system's present is determined by its trajectory — and answer it with two different objects: a finite local jet, and a global weighted integral. Three complementary readings organise the comparison (Figure 3): flatness supplies structural conditions, the Free Energy Principle the objective, and CRR a candidate process layer.

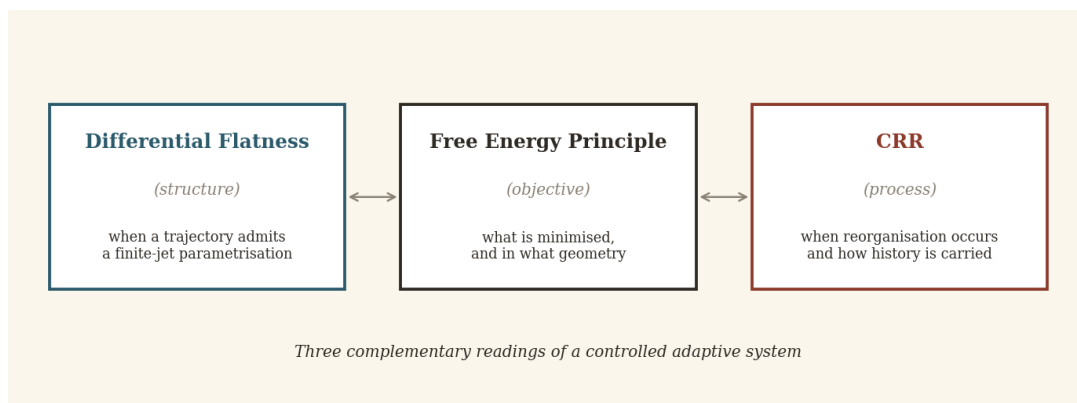


Figure 3. One way to place the frameworks. Flatness characterises when a trajectory admits a finite-jet parametrisation; the Free Energy Principle specifies what is minimised; CRR proposes when reorganisation occurs and how history is carried across it. These are complementary readings rather than competing accounts.

Probe A. The representation of delay

MPF observe that generalised coordinates become inadequate for systems with delays and for high Brunovský index — their Bergman model, with index three, produces incorrect tracking dynamics. The reason is worth stating sharply, because it is exactly where a history integral earns its keep. Reconstructing a delayed value $f(t - \tau)$ from a finite jet at t is a Taylor expansion, and its radius of convergence is set by the signal's spectrum. For a signal of top angular frequency ω the reconstruction is accurate while $\omega\tau$ is small and **diverges once $\omega\tau$ exceeds about π** — a Nyquist-like

wall — and adding derivative orders does not rescue it (Figure 4, left). At $\omega\tau = 3.5$ the second-order jet errs by more than the amplitude of the signal itself. This is the Brunovsky breakdown in miniature: a delay carries information that no finite number of local derivatives can hold.

A memory integral is the right kind of object precisely because it is infinite-dimensional. A pure delay is a shift, $\exp(-s\tau)$ in the Laplace domain, and the Gamma / Laguerre memory kernels — of the form $\exp(-s/T) \cdot s^{k-1}$ — are **complete in $L^2(0, \infty)$** (a classical result of Szegő). A finite sum of them therefore approximates any delay kernel to arbitrary L^2 accuracy, and the error falls monotonically as kernels are added (Figure 4, right). The convergence is not fast — approximating a sharp delay with smooth kernels is intrinsically demanding — but it is guaranteed, and it is the honest content of the claim: not that CRR represents delay efficiently, but that a history-weighted integral *can* represent it at all, where a finite jet provably cannot.

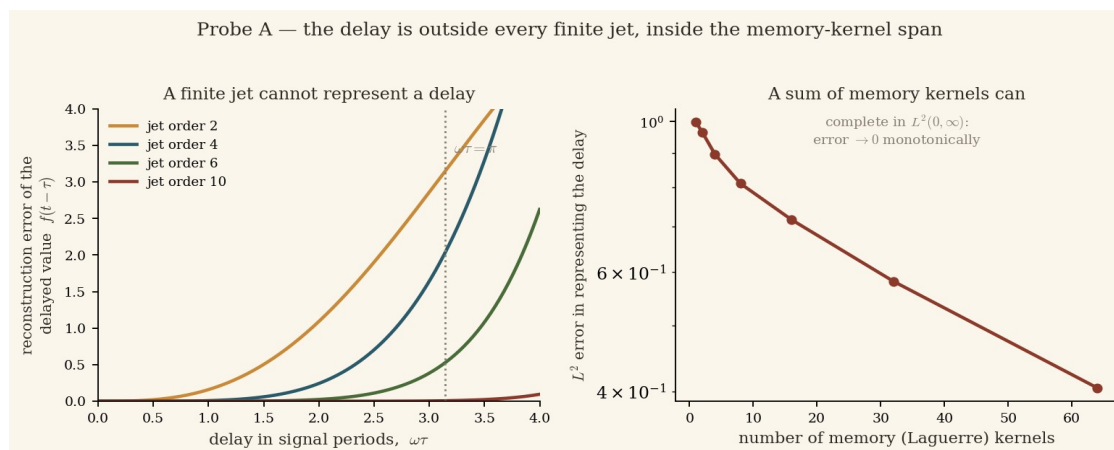


Figure 4. Left: error in reconstructing a delayed value from a finite jet, against delay in signal periods; every order diverges past $\omega\tau \approx \pi$. Right: L^2 error in representing the same delay by a sum of orthonormal Laguerre memory kernels, falling monotonically toward zero by completeness. The two representations differ not in degree but in what they can represent.

The result is a clean representational separation — a delay lies outside the span of every finite jet and inside the closure of the memory-kernel span — and it locates the CRR-flatness relationship exactly. Flatness is the regime in which a trajectory's history collapses to finitely many generalised coordinates; the memory regime, where it does not, is where an integral kernel becomes necessary rather than optional. What this does *not* yet provide is a control law: representing a delay is not the same as designing feedback through it, and that step is left open.

Probe C. Is the flat output the intrinsic clock?

The second probe is more speculative and, if it survives generalisation, more interesting. A differentially flat system has a flat output — the coordinate in which the dynamics become linear under endogenous feedback. CRR has its own privileged coordinate: the Fisher-Rao arc-length parameter, the one in which the coherence clock runs uniformly, $C(\tau) = \tau$. The question is whether these are the same coordinate.

On the Bernoulli manifold the question can be settled by direct calculation, and the answer is yes. With Fisher metric $g(p) = 1/(p(1-p))$, the arc-length coordinate is $\phi = 2 \arcsin \sqrt{p}$, and natural-gradient descent of any potential V — the flow $\dot{p} = -g^{-1}V(p)$ that active inference and CRR both take as basic — becomes, in this coordinate, **ordinary gradient descent**:

$$\dot{\phi} = -dV/d\phi.$$

This identity holds to machine precision at every point tested. The natural gradient, curved in the raw coordinate, is straightened in the arc-length coordinate — which is precisely what it means for that coordinate to linearise the flow. A trajectory confirms the corollary: the accumulated coherence $C(\tau)$ and the arc length travelled in ϕ are the same quantity, so in this coordinate the clock is uniform and the flow is linear at once (Figure 5).

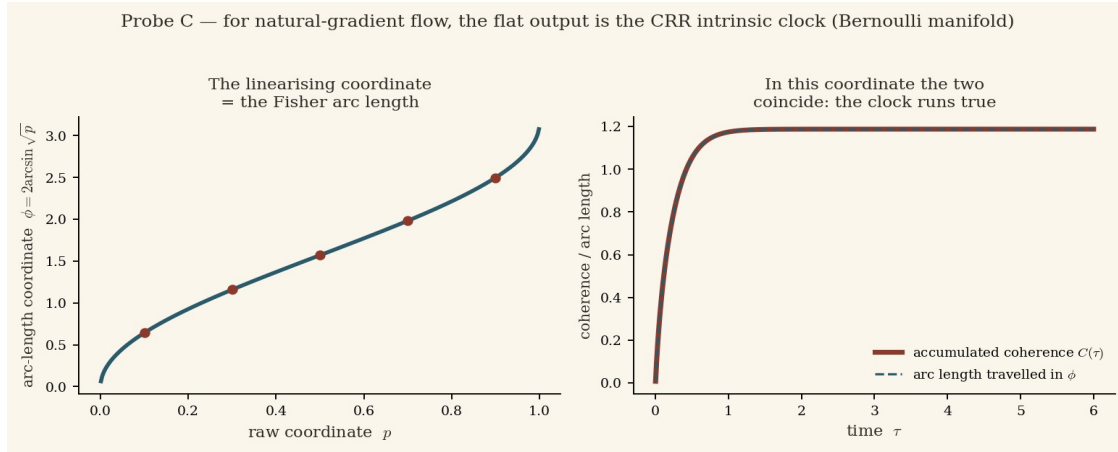


Figure 5. Left: the map from the raw Bernoulli coordinate p to the Fisher arc-length coordinate ϕ , the coordinate that linearises the natural-gradient flow. Right: along a natural-gradient trajectory, accumulated coherence $C(\tau)$ (thick) and arc length travelled in ϕ (dashed) coincide — the coordinate in which the flow is linear is the one in which the CRR clock runs true.

The suggestion, then, is that for natural-gradient flows the differentially-flat output and the CRR intrinsic clock are the same coordinate — that a system being flat and its coherence clock running uniformly are two descriptions of one geometric fact, in the vocabularies of control theory and information geometry respectively. If so, the relationship between the frameworks is not merely analogical; they share a coordinate. Two honest limits attach. First, this has been shown for one manifold, the Bernoulli case, where the natural gradient's linearising coordinate happens to be the arc length; whether it holds for general statistical manifolds, and for the specific flatness conditions MPF require, is unproven and is the obvious next calculation. Second, natural-gradient flow is a special dynamics; a system flat for other reasons need not have its flat output align with any metric coordinate. The claim is therefore a well-posed conjecture with one confirming case.

The bridge, stated once

Both probes turn on the same structural fact, and it is worth naming the bridge explicitly. A finite jet is what a history functional reduces to when the past carries no long-range dependence; the $\exp(C/\Omega)$ kernel describes the regime where it does not. Differential flatness, on this reading, is the short-memory limit in which a trajectory's history collapses to finitely many generalised coordinates, and CRR occupies the complementary regime where that collapse fails — which is exactly the delay-and-high-index regime MPF flag (Figure 6). This is a real and testable idea, and, at present, a conjecture: the theorem that would establish it — a characterisation of when a CRR history-integral admits a finite-jet reduction — is not yet in hand.

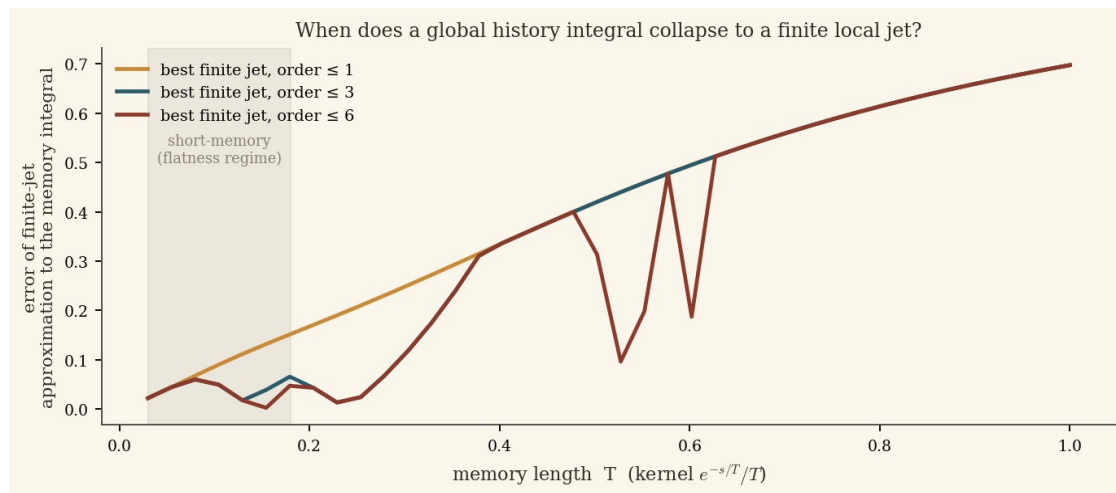


Figure 6. Error of a finite-jet (generalised-coordinate) approximation to a memory integral, as memory length grows. In the short-memory band a low-order jet suffices — the flatness regime; for longer memory the error stays of order one at every order shown — the CRR regime. Computed directly; not fitted.

Discussion and open questions

If there is a synthesis, it is a small one, and perhaps more defensible for being small. Flatness characterises structure — when a trajectory admits an invertible finite-jet parametrisation and hence linearising feedback. The Free Energy Principle specifies the objective. CRR proposes a process account of when reorganisation occurs and how history is carried across it. The genuine bridges are the two probes: a delay lies outside every finite jet but inside the memory-kernel span, and — on the Bernoulli manifold — the flat output and the intrinsic clock coincide. Both were computed rather than asserted, and both point to the same underlying contrast between a finite local jet and a global history integral.

Three questions seem to me the honest residue of the comparison. First, under what conditions does a CRR history-integral admit a finite-jet (flat) reduction — the theorem that would turn the short-memory-limit picture from conjecture into result? Second, does the coincidence of flat output and intrinsic clock generalise beyond the

Bernoulli manifold, and to the specific flatness conditions MPF require? Third, is CRR's fitness for delay and memory systems purely representational, or can the regeneration functional support a genuine tracking law? I offer these to the authors and to the wider community not as extensions already made, but as points of contact worth examining — with the framework's limitations, I hope, stated as plainly as its aspirations.

All figures and identities were computed directly for this paper: the geodesic extents (π and 2π , ratio 2.000000) and the reparameterisation-invariance of the coherence arc length; the finite-jet divergence and Laguerre-kernel L^2 convergence of Probe A; and the flow identity $\dot{\phi} = -dV/d\phi$ of Probe C, verified to machine precision on the Bernoulli manifold with $C(\tau)$ equal to arc length travelled. Nothing is asserted beyond what was computed, and the generalisations are flagged as open. Prepared as a discussion document; the differential-flatness account is that of Mounier, Parr & Friston, Entropy 28(1), 87 (2026).